

Waste2Menu-AI: A Zero-Waste Reverse Recipe and Byproduct Optimization Platform with Community Kitchen and NGO Surplus Dispatch

Hriddho Chatterjee, Jatin Sahu, Binay Kumar Bishwakarma

Department of Computer Science and Engineering, SRM Institute of Science and Technology (SRMIST), Ramapuram, Chennai, India

Abstract

Food waste is an environmental, economic, and social challenge. Waste2Menu-AI is a prototype digital platform designed to address the problem where kitchen byproducts, surplus cooked food, and community demand intersect. The system combines reverse-recipe discovery, scrap recognition, structured byproduct recipes, surplus handling, and a non-commercial community-kitchen/NGO dispatch workflow. Home chefs can identify usable peels, stems, seeds, rinds, and leftovers and obtain recipe guidance; master chefs can author and publish upcycling recipes; and community kitchens can broadcast surplus meals to participating shelters and NGOs for controlled pickup. The prototype is implemented as a web application with a React/TypeScript front end and a Python/FastAPI-oriented service architecture, with relational records for inventory, recipes, orders, NGOs, redistribution batches, and sustainability events. Public dashboard figures are treated as prototype/demo metrics, not independently audited experimental results. The report therefore emphasizes reproducibility, provenance, human oversight, food-safety controls, and a measurable evaluation framework.

Keywords: food waste reduction, reverse recipe recommendation, byproduct valorization, computer vision, community kitchens, NGO food redistribution, circular food systems, sustainability analytics

1. Introduction

Food waste is a major systems problem because resources are consumed to produce, transport, prepare, and store food that may ultimately be discarded. The United Nations Environment Programme (UNEP) estimated that 1.05 billion tonnes of food were wasted in 2022 across households, food service, and retail, representing about 19% of food available to consumers. Households accounted for about 60%, food service 28%, and retail 12%. [1], [2]

Conventional food-management software often focuses on inventory tracking, expiry monitoring, demand forecasting, or disposal logging. Waste2Menu-AI approaches the problem from another direction: instead of treating every kitchen byproduct as waste, it attempts to identify material that may have culinary value and map it to a suitable recipe, preparation workflow, or community redistribution pathway. This is the central reverse-recipe concept.

The platform has three principal user groups. Home chefs receive guidance for transforming kitchen scraps and leftovers into dishes. Master chefs can contribute recipes and culinary knowledge. NGOs, shelters, volunteers, and community kitchens provide the social-distribution layer for surplus prepared food. The intended digital pathway is: identify -> transform -> prepare -> redistribute -> record impact.

2. Problem Statement and Objectives

The project addresses four related problems: usable food byproducts may be discarded because people do not know how to use them; recipe discovery is normally ingredient-forward rather than byproduct-forward; surplus cooked food may not reach a nearby organization quickly enough; and sustainability claims are difficult to verify when underlying events are not recorded systematically.

Objective	Description
Byproduct recognition	Identify or manually register kitchen byproducts and usable leftovers.
Reverse recipe generation	Map byproducts to suitable recipes and preparation steps.
Recipe knowledge sharing	Allow chefs to author, preserve, and publish zero-waste recipes.
Surplus dispatch	Broadcast surplus cooked meals to nearby participating shelters/NGOs.

Objective	Description
Traceable handover	Use claim records and an OTP-based handover step.
Impact accounting	Record diverted quantities and sustainability events for later audit.

Table 1. Project objectives.

3. Related Work

Prior research has explored food recommendation using inventory information and sensing technologies. One representative approach combines household inventory management, cameras, load cells, and deep-learning-based recognition to recommend recipes from available ingredients. Such systems demonstrate the value of connecting sensing and recipe recommendation, while Waste2Menu-AI focuses specifically on kitchen byproducts and surplus community redistribution.

Research on fruit and vegetable byproducts also establishes the rationale for valorization: residues can contain nutrients and functional compounds and can sometimes be recovered instead of being discarded. Waste2Menu-AI translates this principle into a user-facing workflow by starting from the byproduct and searching for an appropriate culinary use.

The main distinction is therefore the integration of reverse-recipe discovery, community recipe authoring, surplus meal broadcasting, NGO/shelter claiming, and sustainability event logging in one prototype. Recipe recommendation and image recognition are not claimed as novel individually.

4. System Overview

The public prototype provides Home & Impact, Ingredients Catalog, NGO Dispatch, Home Chef, recipe authoring, and authentication. The Home Chef workflow accepts a photograph or selected scrap sample, presents detected ingredients, and offers recipe filters. The Master Chef workflow captures byproduct, course, dietary preference, author, affiliation/tradition, preparation time, difficulty, servings, pantry staples, steps, and chef tips.

The community module provides a non-commercial surplus broadcast. A kitchen can register the meal name, prepared-by information, portions, dietary classification, ready time and temperature, pickup location, contact information, packaging notes, and dietary notes. A participating NGO or shelter can claim a batch and receive a six-digit handover OTP.

5. Proposed Reverse-Recipe and AI Pipeline

Stage	Input	Processing	Output
1. Capture	Photo / manual entry / weight	Create event	Normalized item record
2. Recognition	Kitchen image	Ingredient/scrap classification	Candidate labels
3. Validation	Candidate labels	Human confirmation	Approved items
4. Retrieval	Approved byproducts	Filter/rank recipes	Recipe candidates
5. Preparation	Selected recipe	Steps, quantities, servings	Cookable plan
6. Routing	Prepared excess	Broadcast and claim matching	Pickup batch
7. Ledger	Event quantities	Apply documented factors	Impact record

Table 2. End-to-end processing pipeline.

5.1 Computer Vision

The architecture is compatible with a YOLOv8-class object-detection workflow for recognizing visible ingredients or scraps. Model output should be treated as a candidate list rather than a food-safety decision. Production use should expose confidence scores, allow user correction, and log the final human-confirmed class.

5.2 Reverse Recipe Retrieval

Let S be the set of confirmed byproducts and R the recipe catalog. A compatibility score can be defined as $\text{Score}(r, S) = w_1 * M + w_2 * D + w_3 * C + w_4 * P$, where M is byproduct match, D dietary compatibility, C course

compatibility, and P pantry feasibility. The weights should be calibrated using labelled evaluation data rather than assumed to prove accuracy.

5.3 Optimization Layer

For larger kitchens, recipe selection can be extended to an integer-linear-programming formulation. Decision variables can represent portions of candidate dishes; constraints can represent available byproduct mass, recipe yield, preparation capacity, serving demand, and dietary requirements. The objective can combine waste diversion and operational value subject to documented constraints.

6. NGO and Community Kitchen Workflow

The NGO module is a central part of the project. Its purpose is to provide a short path from surplus preparation to community pickup: (1) kitchen creates a surplus batch; (2) quantity, dietary information, readiness time, temperature/packaging notes and location are recorded; (3) eligible organizations see the batch; (4) an organization claims it; and (5) handover is confirmed using an OTP.

Geographic matching can use Haversine distance between a kitchen and registered organizations. Distance alone, however, must not determine eligibility. Production deployment should include organization verification, food-holding limits, allergen disclosure, packaging integrity, maximum dispatch windows, pickup capacity, and applicable local food-safety and charity requirements.

7. Implementation Architecture

```
[ Home Chef / Master Chef / NGO Web UI ]
|
[ React + TypeScript ]
|
[ FastAPI Services ]
/ | \
[Vision] [Recipe/Opt.] [NGO Dispatch]
| | |
[Scrap] [Recipe DB] [Claim + OTP]
\ | /
[Sustainability Ledger]
|
[SQLite / PostgreSQL]
```

Figure 1. Logical architecture of the prototype.

The described prototype stack includes React 18, TypeScript, Vite, Tailwind CSS, Framer Motion and Zustand on the front end; Python/FastAPI, Pydantic and SQLAlchemy-oriented services on the back end; SQLite for prototype storage with PostgreSQL as a deployment option; and optimization components compatible with SciPy/PuLP/CBC-style solving.

8. Data Model

Entity	Purpose
User	Authentication, role and affiliation.
RawInventory	Available ingredients or raw inventory records.
ScrapLedger	Byproduct quantity, category and diversion event.
ByproductRecipe	Recipe definition linked to a byproduct.
DynamicSku	Temporary/surplus product representation.
Order	Prepared or purchased item transaction record.
NGO	Registered shelter/community organization.
RedistributionBatch	Surplus meal claim and handover record.
SustainabilityLedger	Calculated impact event with provenance.

Table 3. Core logical data entities.

9. Sustainability Accounting

The prototype includes indicators for diverted mass, greenhouse-gas avoidance, meals supported, and water savings. A project-level equation can be written as $\text{CO2e_avoided} = \text{diverted_mass_kg} \times \text{emission_factor_kgCO2e_per_kg}$. The emission factor must be documented, versioned, and appropriate to

the waste type and disposal pathway. The same principle applies to water or meal-equivalent calculations. The public website displays dashboard figures including 12,450+ kg organic kitchen scraps saved, 31,125 kg CO2e avoided, 1,420+ recipes, and 8,600+ meals dispatched. These are treated here as prototype/dashboard metrics, not independently audited results. They should only be presented as experimental evidence after event-level records, measurement boundaries, conversion factors, and validation are documented.

10. Evaluation Methodology

Component	Metric	Evaluation
Vision	Precision, recall, F1, mAP	Held-out annotated images; per-class results.
Recipe retrieval	Precision@k, Recall@k, nDCG	Expert-labelled relevance set.
Usability	SUS / task completion	Representative users from all three roles.
Waste diversion	kg diverted / baseline	Before-after or controlled kitchen study.
NGO dispatch	Claim rate, pickup time, failures	Timestamped pilot batches.
Impact accounting	Ledger reproducibility	Recompute every dashboard value from events.

Table 4. Reproducible evaluation plan.

A publication-quality experimental study should report dataset size, train/validation/test splits, annotation protocol, hardware, software versions, hyperparameters, baselines, confidence intervals where appropriate, and statistical tests for before-after comparisons. Without these details, numerical claims should remain explicitly labelled as prototype or demonstration results.

11. Security, Privacy, and Food-Safety Considerations

Authentication: role-based access should prevent unauthorized recipe publishing, batch claiming, or administrative actions.

Personal data: phone numbers and organization details should be minimized, protected, and retained only as necessary.

OTP security: handover codes should be short-lived, single-use, rate-limited, and securely stored.

Food safety: image recognition cannot establish edibility and must never be presented as a safety certification.

Allergens: recipes and surplus batches should support structured allergen declarations and dietary labels.

Auditability: every sustainability figure should trace to a timestamped source event and versioned conversion factor.

12. Limitations

- 1) The public prototype is primarily a software demonstration and does not by itself establish real-world waste-reduction causality.
- 2) Computer-vision performance depends on training data, lighting, occlusion, camera angle, and ingredient diversity.
- 3) Recipe compatibility is culturally and nutritionally context-dependent and should support human review.
- 4) NGO matching based on geography alone is insufficient for food-safety, capacity, legal, and operational constraints.
- 5) The public interface contains a catalog-count inconsistency: some areas refer to 30 items while another section refers to 35 ingredients. The source of truth should be unified before publication.
- 6) Environmental impact figures depend on system boundaries and conversion factors; a dashboard number is not a lifecycle assessment.

13. Future Work

Future development can focus on a larger annotated scrap-image dataset, multimodal recognition, personalized recipe ranking, nutrition and allergen reasoning, demand-aware surplus routing, route optimization, expiry-aware scheduling, organization verification, multilingual interfaces, offline-first NGO workflows, and event-level impact auditing. A future study should compare Waste2Menu-AI against ingredient-only recommendation baselines and measure the effect of the reverse-recipe workflow on actual kitchen waste.

14. Conclusion

Waste2Menu-AI presents an integrated approach in which kitchen byproducts are treated as potential inputs to a reverse-recipe system and surplus cooked food is connected to a community redistribution workflow. Its technical contribution is the integration of computer-vision-assisted identification, recipe knowledge, optimization-oriented selection, role-based workflows, NGO batch claiming, and sustainability event logging. Its social contribution is the explicit inclusion of community kitchens and shelters in the same workflow. The appropriate next step toward a strong scholarly publication is controlled validation with transparent datasets, documented conversion factors, usability studies, and safety/compliance procedures.

References

- [1] United Nations Environment Programme, "Food Waste Index Report 2024," UNEP, 2024. Available: <https://www.unep.org/resources/publication/food-waste-index-report-2024>
- [2] United Nations Environment Programme, "Food Waste Index Report 2024: Key Messages," UNEP, 2024. Available: <https://wedocs.unep.org/handle/20.500.11822/45275>
- [3] United Nations, "Sustainable Development Goal 12: Responsible Consumption and Production," Target 12.3. Available: <https://unstats.un.org/sdgs/>
- [4] Food and Agriculture Organization of the United Nations, "Food Loss and Food Waste," FAO. Available: <https://www.fao.org/food-loss-reduction/>
- [5] United Nations Environment Programme, "Towards Zero-Waste and Circular Cities: Key Findings of the Food Waste Index Report 2024," 2024. Available: <https://www.unep.org/events/unep-event/towards-zero-waste-and-circular-cities-key-findings-food-waste-index-report-2024>
- [6] Waste2Menu-AI public prototype, "A Zero-Waste Reverse Recipe & Byproduct Optimization Engine." Available: <https://hriddhochatterjee.github.io/Waste2Menu-AI/>
- [7] Waste2Menu-AI project documentation and implementation materials supplied for this research report.

Notation fix: This version intentionally uses the ASCII notation CO₂e everywhere instead of a Unicode subscript character. This prevents the square-box (CO²e) rendering problem caused by unsupported glyphs in some PDF fonts/viewers.